

Himanshu Ashish Desai

Mechanical and Aerospace Engineer

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Portfolio

EDUCATION

Georgia Institute of Technology

Expected August 2027

Master of Science | Aerospace Engineering | CGPA – 3.660/4.0

Relevant Coursework: Principles of Autonomy, Advanced Aerodynamics, Fracture Mechanics

Vellore Institute of Technology, Vellore

2020 - 2024

Bachelor of Technology | Mechanical Engineering | CGPA – 3.880/4.0

Relevant Coursework: Design Process Planning and Management, Machine Design, CFD

ENGINEERING PROJECTS

Numerical Analysis on Scramjet Combustor (Published in *IJHE: Elsevier*)

January 2024 – August 2024

Dr. Padmanathan P. | Vellore Institute of Technology, Vellore

- Designed and simulated **scramjet** combustors with **diamond strut** injectors in **SOLIDWORKS** and **ANSYS**, validated against experimental data, achieving a **6.43% efficiency improvement** over the baseline model.

Modular Drone Generative Design

July 2023 – October 2023

Dr. S. Senthur Prabu | Vellore Institute of Technology, Vellore

- Designed a **modular 6-axis UAV** with detachable wing architecture using **Fusion 360** to optimize weight, payload adaptability, and flight stability; performed multi-physics analysis, **FMEA**, reducing critical failure risk (**RPN up to 576 → 196**), and improving structural reliability.

Analysis of Cooling Systems for Car Battery Pack

January 2023 – April 2023

Dr. Padmanathan P. | Vellore Institute of Technology, Vellore

- Conducted **CFD** and transient thermal analysis of battery cooling systems in **ANSYS Fluent**, comparing air, water, and ethylene glycol, achieving improved heat dissipation by **17%** with liquid cooling

WORK/INTERNSHIP EXPERIENCE

Reliance Industries Limited (RIL)

(Jamnagar, Gujarat, India)

Shift Field Engineer | JMD DTA – Captive Power Plant (CPP)

August 2024 – May 2025

- Analyzed performance and reliability of **Gas Turbines (Frame 9E/6B)**, **HRSBs**, and **Steam Turbine Generators** in a **276 MW** captive power plant, identifying efficiency losses and operational deviations.
- Performed root-cause analysis of GTs, HRSBs, and STGs, supporting diagnostics, inspections, and commissioning of a **400 MW** expansion.

Turnaround Maintenance Engineer | JMD DTA – Crude Distillation Unit – 1 (CDU-1)

March 2025 – April 2025

- Assessed maintenance needs of **19 heat exchangers** and **2 crude heaters**, supporting tube bundle extraction and heat transfer restoration
- Executed shutdown activities using **SAP PM/MM**, ensuring timely maintenance and adherence to safety targets.

Summer Intern | JMD DTA – Central Engineering Services (CES)

May 2023 – July 2023

- Performed **condition monitoring** on a **2.5 MW blower**, diagnosing the root cause of frequent trips and **>90 dB noise** levels.
- Pinpointed faulty drive-end bearing, suggested **corrective actions**, extending bearing life, and **reducing maintenance downtime**.

Associated Power Structures Limited

(Vadodara, Gujarat, India)

Internship | Vocational Trainee

June 2022 – July 2022

- Assisted in surveys and design of **2 transmission towers**, covering **welding** and **galvanizing** processes, ensuring safety standards.
- Learned full tower lifecycle, quality control measures, and compliance with 765 kV load-bearing standards.

Shakun Polymers Limited (Orbia Group)

(Halol, Gujarat, India)

Summer Intern | Manufacturing

May 2022 – June 2022

- Analysed the **Bühler palletizing system** in a high-volume polymer plant producing over **50 metric tons per day**.
- Gained technical expertise in compounding workflows, material evaluation techniques, and cable-grade polymer technology.

SKILLS AND COMPETENCIES

Technical Skills – CAD/CAM/CAE, GD&T, Finite Element Analysis (FEA), CFD, Thermal Fluid Systems, Mechanical Maintenance and Operations, Machining, FDM, 3D Printing, Materials Testing, Lean Six Sigma

Software Tools – AutoCAD, Fusion 360, SOLIDWORKS, XFOIL, ANSYS (Fluent, Mechanical, ICEM CFD), SAP (PM, MM), Microsoft Suite.

Management Courses Taken – Total Quality Management & Reliability (QFD, FMEA), Industrial Engineering Management, Supply Chain Management, Lean Manufacturing Technology (Kaizen, JIT, VSM, TPS, Gemba Walks, Poka-Yoke, SMED)

Programming Languages – MATLAB, Simulink, Java (OOPS), C++ (basics), HTML/CSS/JS (Front End Dev.)

AWARDS AND CERTIFICATIONS

Autodesk CAD/CAM/CAE for Mechanical Engineering by AUTODESK

April 2023

MATLAB Programming Specialization by Vanderbilt University

May 2023

Six Sigma Green Belt Specialization by University Systems of Georgia

May 2023

Certified SOLIDWORKS Associate (CSWA) by Dassault Systems

August 2023